

1. The "Why": Project Objectives

I took on this project to solve a classic retail business question: "**What actually drives sales?**" Is it promotions? Seasonality? Or external economic factors like CPI and unemployment? My goal was to move beyond just calculating numbers and extract actionable insights that store managers can use to make data-driven decisions.

2. The Data Journey (Cleaning & Prep)

The data wasn't "clean" out of the box. I took a hands-on approach to prepare it:

- Converted the Date column to datetime to unlock time-based analysis.
- Ensured data integrity by handling duplicates and verifying consistency.
- Performed **Feature Engineering** by creating a Year column, allowing for year-over-year performance comparisons.

3. Key Findings (The "So What?")

After digging into the data, here is what stood out:

- **The Powerhouses:** Stores 20, 4, and 14 are absolute performance leaders, consistently driving the highest sales volume.
- **The Holiday Spike:** Holidays clearly trigger a massive boost in sales. This is a golden opportunity for marketing and inventory teams to double down.
- **The Insightful Surprise:** Contrary to my initial hypothesis, economic indicators like unemployment and fuel prices showed a weak correlation with sales. This suggests that **internal operational factors** are significantly more important than external economic shifts for Walmart's sales performance in this dataset.

4. The Dashboard (Power BI)

I built an interactive dashboard to turn these findings into a visual narrative. It enables stakeholders to:

- Monitor core sales KPIs at a glance.
- Compare performance across different branches instantly.
- Visualize holiday impact trends to optimize future planning.

5. My Recommendations for Management

If I were advising the business, I would suggest:

1. **Benchmarking:** Conduct a "Case Study" on Store 20 to identify their operational best practices, then attempt to replicate that "recipe for success" in lower-performing branches.
2. **Proactive Inventory:** Use the holiday sales patterns to optimize stock levels well in advance of peak seasons.
3. **Future-Proofing Analysis:** To explain the sales fluctuations that aren't tied to economic factors, we need to dive deeper into product categories, customer segmentation, and marketing campaign data.

6. Conclusion

This project was a great exercise in bridging the gap between Python-based data analysis and Power BI-based reporting. It provided a full-cycle view of performance and highlighted that, in this specific context, operational excellence beats economic cycles.